Medical Insurance Charges: Sex-Based Analysis

Matthew deHaas, Callum Mackenzie, Saksham Saksham, ...

Course: STAT 306

Instructor: Kenny Chiu

March 24, 2026

Contents

1 Introduction **2**

2 Analysis **2**

2.1 Univariate Summaries 2

2.2 Bivariate Relationship 3

2.3 Confounders 4

2.4 The Case for Regression 5

3 Conclusion **6**

4 Statement of AI **6**

5 Appendix **6**

1. Introduction

The following report focuses on whether sex exerts predicative power on medical insurance premiums after controlling for body mass index (BMI) and age. Motivation comes from a collective interest in whether non-meritorious, sex-based discrimination is present in the context of medical insurance applicants. Such an influence could potentially suggest that insurance institutions operate under systemically discriminatory practices as it pertains to sex. Data was observed from a collection of studies conducted by accredited institutions such as the World Health Organization, the National Institute of Health, the Centre for Disease Control and Prevention, and more. The dataset itself provides information on approximately 100,000, including their "...demographics, socioeconomic status, health conditions, lifestyle factors, insurance plans, and medical expenditures". As mentioned, age and BMI will serve as a primary basis for analysis. It may be helpful to note that the source under scrutiny that collected these studies underwent a partial cleaning and normalization procedure, so certain pieces of data, particularly in the extremities, may seem strange in isolation.

2. Analysis

2.1. Univariate Summaries

To begin, we introduce the variables used for analysis and their respective shapes.

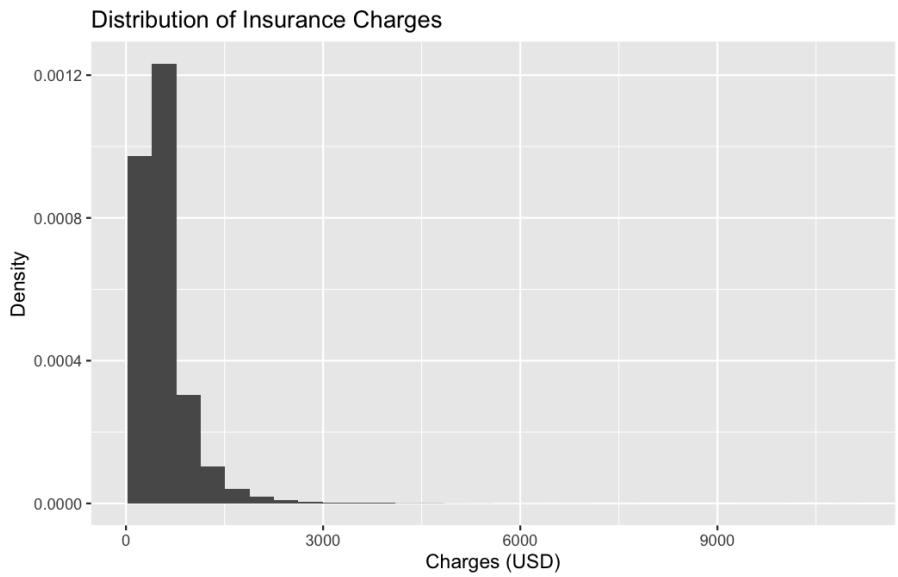


Figure 1. Distribution of annual insurance premiums. There is significant right skew, so perhaps a log transformation is more appropriate.

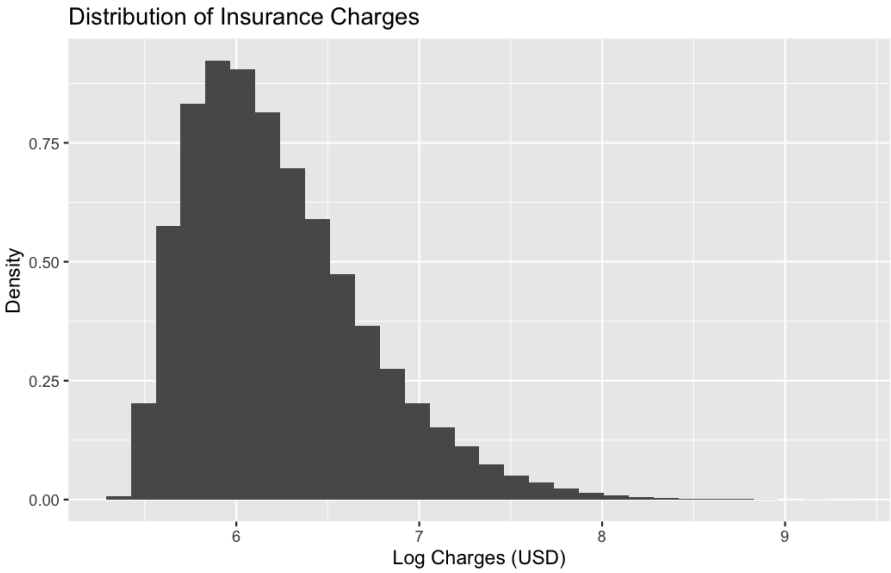
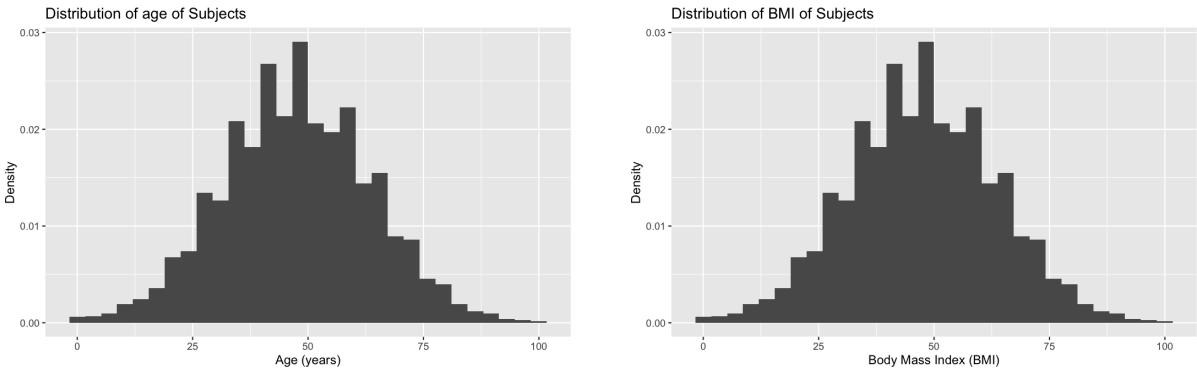


Figure 2. Much more suitable for analysis.



(a) Age **(b)** BMI

Figure 3. Distributions of BMI and Age. We observe that they both seem are roughly normal.

2.2. Bivariate Relationship

We now explore how sex in isolation relates to insurance premiums.

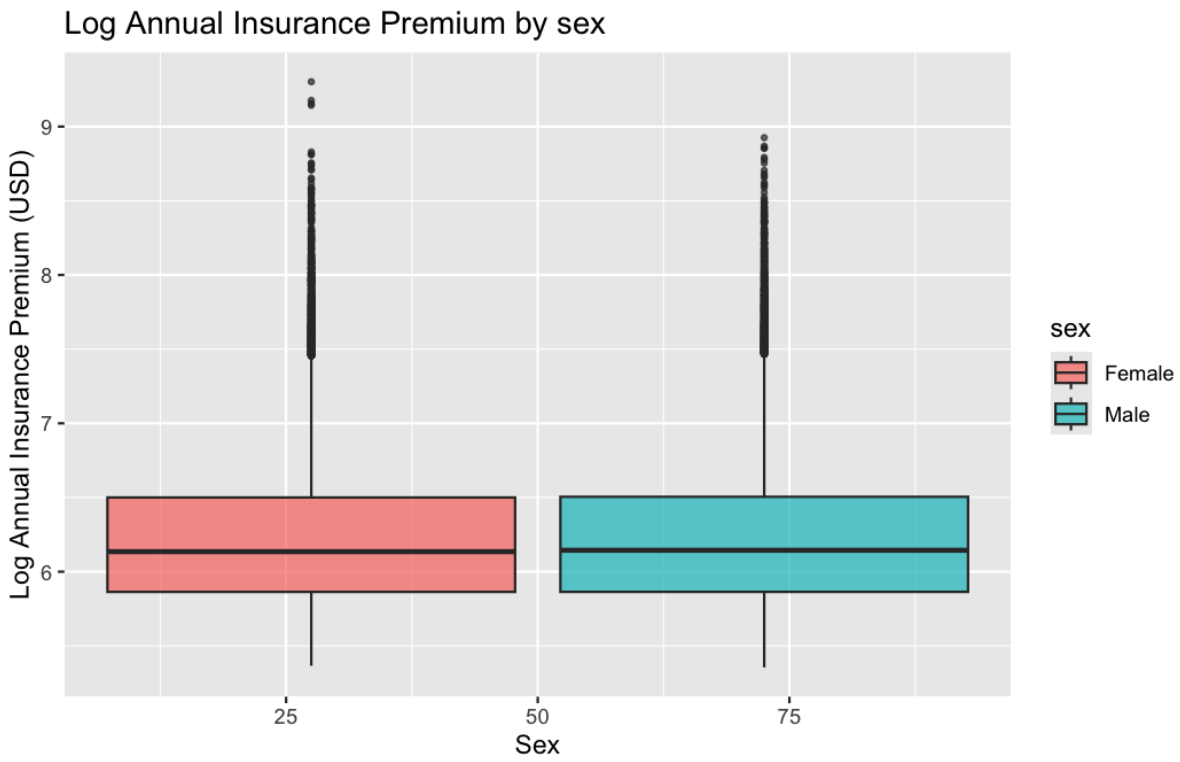
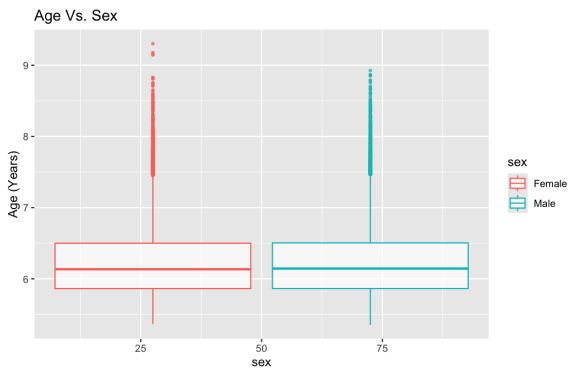


Figure 4. Log Insurance Premiums vs. Sex

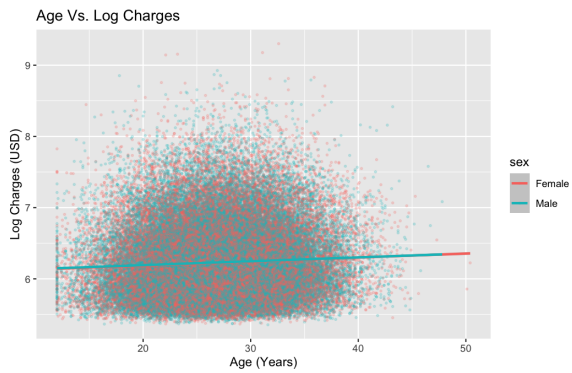
Figure 5. Insurance premiums as a function of the sexes

2.3. Confounders

This looks promising, but a simple male to female comparison is naive. We now consider how age and BMI are distributed across the sexes, as well as how they compare to (log) insurance premiums.

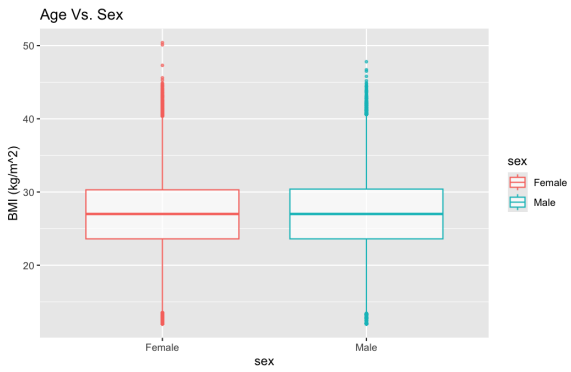


(a) Age vs. Sex

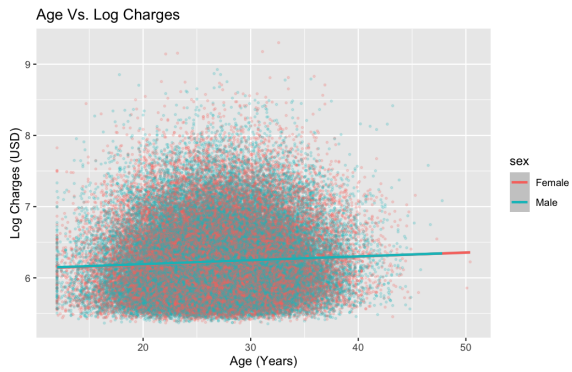


(b) Age vs. Log Insurance Premiums

Observe that the distribution of age among the sexes is nearly identical as per the box plots. In addition, the relationship between age and age and premiums is weakly positive among the sexes. We can infer that age is an unlikely influence on the differences in premiums among the sexes.



(a) BMI vs. Sex



(b) BMI vs. Log Insurance Premiums

Similarly, BMI itself and its relationship with premiums are weakly positive among the sexes. We can comfortably conclude now that the variables age and BMI under scrutiny here are not likely to be confounding in comparison between sex and medical insurance premiums.

2.4. The Case for Regression

There are myriad factors to consider in attempting to understand the prices of medical insurance premiums, especially with the contentious notion of systemic discrimination. Thus, we look to linear regression to explain whether a sex is a primary determinant of what one is charged for medical insurance.

Table 1. Staged Linear Regression Results

	Dependent variable:	
	log(annual_premium)	
	Sex Model	Sex + Age + BMI
Sex	0.0046	0.0042
	(0.0031)	(0.0031)
Age (Years)		0.0048***
		(0.0001)
BMI (kg/m^2)		0.0055***
		(0.0003)
Constant	6.2260***	5.8520***
	(0.0022)	(0.0096)
Observations	97,987	97,987
R^2	0.00002	0.0278
Adjusted R^2	0.00001	0.0278
Residual Std. Error	0.4846 (df = 97985)	0.4778 (df = 97983)
F Statistic	2.2422 (df = 1; 97985)	935.4822*** (df = 3; 97983)
Note:		*p<0.05; **p<0.01; ***p<0.001

We now have a clear picture of the data. First, the sex coefficient is not a significant contributor to the response variable alone, evident by the p-value not reaching 0.05. This is consistent with the visual intuition obtained from the previous scatter plot that related age and bmi to log premiums separately. This is strengthened by the R^2, the variable responsible for the proportion of variance in log premiums explained by the covariates in the model. It was at a negligible 0.00002 in the sex mode, while jumping to 0.0278 in the second one.

3. Conclusion

The motivating question for this report was whether sex was a significant predictor for medical insurance premiums when controlling for demographic conditions such as age and BMI. According to this study, we are confident in the assertion that there exists no strong relationship between what an individual is charged and their sex. However, the particular covariates used only account for under 3% of the variance in insurance premiums, thus there may still exist unobserved cofounding variables.

4. Statement of AI

5. Appendix